

UI and UX first principles

Study guide

A companion to the session and the Fig & Filter build exercise

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This guide carries the detail the slides leave out. Read it after the session to consolidate, or during the build when you need to check a rule. Nothing in the exercise requires knowledge beyond this document.

1. What UI and UX actually are

UI is what the product looks and behaves like on the screen. UX is what it feels like to live with the product — everything that happens before, during, and after you touch the screen.

The term user experience was coined by the cognitive scientist Don Norman in the mid-1990s while he was at Apple. He needed a phrase that covered more than the interface, because his role spanned industrial design, manuals, marketing, and support — all of which shape how a customer feels about a product. The Nielsen Norman Group, which Norman co-founded with Jakob Nielsen, still defines user experience as "all aspects of the end-user's interaction with the company, its services, and its products."

User interface design is narrower: the craft of the visible and interactive layer — buttons, layout, typography, colour, motion, the flow from one screen to the next. UI is to UX what the dashboard is to driving a car. The dashboard matters, but a beautiful dashboard cannot save you from an engine that will not start.

An everyday illustration: a café's website has beautiful photographs, an elegant menu, and clearly listed opening hours — but the hours are from last year, and you travel there on a Monday to find the café closed. The screen was excellent, and the experience still failed. Norman and Nielsen make the same point with a film-review website whose interface is flawless but whose database only lists big-studio releases: a user searching for an independent film meets an excellent interface and a broken experience at the same time.

Why this matters today specifically: in this session an artificial intelligence (AI) assistant produces your user interface. It will be fast and reasonably attractive by default. What the AI cannot know is your user — who arrives at each page, what they need, and what would make them trust a small café enough to walk in. That judgment is the user-experience layer, and it stays with you. Your job in the build exercise is not to write HTML (HyperText Markup Language) code by hand; it is to make sure the code the AI writes serves the right experience.

2. The five questions

Design principles are easiest to use when phrased as questions you can ask of any screen. These five come from some of the most-replicated findings in human-computer interaction, and they were chosen because they predict the typical faults of AI-generated websites: everything equally loud, too many calls to action, unconventional navigation, and tap targets sized for a mouse.

Question 1 — What do I see in the first three seconds? (visual hierarchy)

Users scan; they do not read. Nielsen Norman Group's eye-tracking studies (first published 2006, replicated since) show readers scanning web pages in an F-shaped pattern — across the top, a shorter sweep below, then down the left edge. Whatever sits where the eye lands first is your message, whether you intended it or not.

Hierarchy is built from three levers: size (bigger reads as more important), position (top and left get seen first in left-to-right languages), and contrast (the element that differs from its surroundings draws the eye — the Von Restorff effect, first documented in 1933). The test: squint at the page, or view it at arm's length on a phone. One element should be unmistakably the most prominent, and it should be the element that serves the page's job. When every element is emphasised, the reader cannot tell which one matters.

Question 2 — What is this page for? (one job per page)

Every page gets exactly one job, stated in one sentence, with one primary call to action that matches it. The Fig & Filter spec is written this way deliberately: the home page's job is make a stranger decide to walk in; the catering page's job is get an office manager to send an enquiry email. Secondary links can exist, but they are visually demoted — one button looks like a button; the rest look like links.

If you cannot state the job in one sentence, the page will fail regardless of how good it looks, because every element on it will compete instead of cooperate.

Question 3 — Where does this surprise the user? (Jakob's Law)

"Users spend most of their time on other sites. They prefer your site to work the same way as all the other sites they already know." — Jakob Nielsen, 2000

Your users have spent thousands of hours on other websites and have built deep, unconscious expectations: logo top-left and clickable to home, navigation across the top, contact details in the footer, links that look like links. Every violated expectation spends the user's attention on something that produces no benefit — they must stop, work out

where the thing is, and resume the task they came for.

For almost every decision you will make today, the familiar choice serves the user better than the novel one. This is not an argument against creativity; it is an argument for spending your creativity where it pays (the photography, the voice of the copy, the food) and following convention in navigation and structure. A consistent pattern, repeated, stops drawing attention to itself — and navigation works best when it draws no attention at all.

Question 4 — What can be removed? (Hick's Law)

Hick's Law (Hick and Hyman, 1952): the time it takes to make a decision grows with the number of choices presented. The formula itself does not matter for our purposes; the shape of the relationship does — going from two choices to four roughly doubles decision time.

In practice: identify the one or two things you want the visitor to do and visually demote the rest; group related options so the user makes two small decisions instead of one large one; and hide genuinely secondary material behind a "more" pattern. A home page with one clear call to action almost always outperforms one with seven equally weighted links. When you review your AI-generated pages, this is the question that finds the most fixes — AI assistants love to add sections.

Question 5 — Can a thumb hit this? (Fitts's Law)

Fitts's Law (Paul Fitts, 1954) is the most-cited law in human-computer interaction: the time to reach a target depends on its distance and its size. A closer target is faster to reach, and a bigger target is faster to hit. The law was validated for arm movement and has held for mouse cursors, finger taps, and styluses.

Practical rules for today: make important targets large — the Web Content Accessibility Guidelines (WCAG) version 2.2 set a minimum touch target of 24×24 pixels (px), and the widely recommended size is 44×44 px, because fingers are much larger than cursors; put targets near where the thumb already is; and remember that screen edges are the easiest targets of all, because you cannot overshoot them — which is why mobile navigation lives at the very top or very bottom of the screen.

3. Mindset one: a website is not an app

The same five questions apply to both, but the weightings are opposite, because the person on the other side is in a different situation.

A website serves a visitor. A stranger arrives — usually from a search or a recommendation — and is deciding whether to trust you and act. Their journey is: arrive › orient › believe › act. Orient: where am I, is this what I searched for? Believe: does this place look real, current, and worth my time — photos, prices, hours, a story? Act: walk in, email, call. Content hierarchy and credibility cues do the work. The visitor may come once and never return; the site must succeed on the first visit.

An app serves a returning user. Someone who has already chosen you opens the app to complete a task, often daily. Their journey is a workflow, repeated until it becomes habit. Speed, remembered state, and consistency do the work; every novelty is a cost paid on every single visit. Think of the difference between an airline's marketing site (persuade a stranger to book) and its check-in app (get a known passenger a boarding pass in under a minute) — same company, same brand, opposite designs, because the mindset of the person holding the phone is opposite.

	Website	App
Who arrives	A stranger, deciding	A returning user, doing
Frequency	Once, or rarely	Daily or weekly
The page's job	Persuade and inform	Complete a task
What wins	Clarity and trust	Speed and habit
Failure mode	Visitor doesn't believe, leaves	Task takes too long, user churns

Today you are building a website. Fig & Filter's visitor is a stranger on a phone deciding whether to walk in, and an office manager on a laptop deciding whether to email.

Persuading that visitor matters more than adding features: the site needs no logins and no ordering system — it needs clear hierarchy, good photography, accurate hours, and one clear action per page.

4. Mindset two: mobile and laptop are different bodies

The same site will be used by two different physical situations, and responsive design is the discipline of serving both.

The input is different. A mouse pointer is precise to a few pixels; a thumb covers roughly a centimetre of glass and hides what it touches. This is where the 44 px target rule comes from, along with generous spacing between links and the rule against interactions that depend on hovering — there is no hover on a touchscreen.

The canvas is different. A laptop viewport (the visible area of the browser window) is wide — commonly 1280 px and up — and supports columns side by side. A phone is one narrow column, commonly 375 px wide. A layout serves the narrow screen by restacking: columns become a vertical sequence, the navigation collapses behind a toggle, and images crop rather than shrink. Body text stays at 16 px or larger; if a visitor must pinch-zoom to read, the page has failed.

The person is different. The laptop user is seated and focused. The phone user may be standing on a street, one-handed, in sunlight, with three seconds of patience. Designing for mobile means serving the same content to a person who has less time, less precision, and less screen.

Design the phone view first. Lay out the 375 px view before the 1280 px view. The narrow screen forces you to rank every element by importance, and the elements you keep are the ones that matter most. The wide screen then becomes the easy problem: the same ranked content with more room to breathe. This is why the exercise checklist tests your live site at 375 px on every page — the mobile view is where design judgment shows.

5. From principles to instructions

You will not draw the screens or write the HTML. Your leverage is in two documents — the `DESIGN.md` you give the AI, and the review you run on what comes back.

What a good `DESIGN.md` contains. Four things per site: the audience and the one job of each page (lifted from the spec, in your own words); the hierarchy of each page — what is first, what is biggest, what is the single primary action; the breakpoint behaviour — what collapses, what stacks, what survives at 375px; and the brand tokens — the exact colours, fonts, and voice from `brand.md`, so the AI does not invent its own.

Write instructions the AI can be tested against. "Make it clean and modern" gives the AI nothing definite to obey and gives you nothing to check afterwards. A testable instruction states a condition you can verify by looking at the finished site:

Vague instruction	Testable instruction
"Make it clean and modern"	"One primary call to action per page; everything else styled as plain links"
"Should work on mobile"	"Navigation collapses below 768 px; no horizontal scrolling at 375 px"
"Easy to read"	"Body text minimum 16 px; line length 60–75 characters on desktop"

Vague instruction	Testable instruction
"Make the buttons nice"	"All tap targets at least 44×44 px with 8 px spacing"

Every row on the right can be verified with your own eyes on the live site, which means the AI's output can be checked against it too. When a generated page fails a check, quote the instruction back to the AI and ask for the fix — do not start the prompt from scratch.

Judging a Stitch layout. Run the five questions against the layout before you build it. It is five minutes' work to reject a bad layout at the Stitch stage and half an hour's work to fix it after the HTML exists.

The review loop. After deploying: open the live URL (web address) on your phone and your laptop, run `checklist.md` on every page, fix the worst failures, and deploy again. Publishing once and never checking is how broken sites reach the public, so plan for at least one full review-and-fix cycle before you submit.

6. Glossary

Term	Meaning
UI (user interface)	The visible, interactive layer of a product: layout, buttons, type, colour, motion
UX (user experience)	Everything about the encounter with a product and its maker, on screen and off
Visual hierarchy	The order in which a design asks to be read, built from size, position, and contrast
Call to action (CTA)	The element asking the user to do the page's one job
Jakob's Law	Users expect your site to work like the sites they already know
Hick's Law	Decision time grows with the number of choices
Fitts's Law	Bigger and closer targets are faster to hit
F-pattern	The scanning path readers follow on text-heavy pages
Breakpoint	The viewport width at which a layout changes (e.g. navigation collapses below 768px)
Responsive design	One site restacking itself to serve every screen width
Viewport	The visible area of the browser; 375px wide is the standard phone test

Term	Meaning
WCAG	Web Content Accessibility Guidelines — the accessibility standard; v2.2 sets the 24px minimum target size
DESIGN.md	The design-intent document you write and hand to the AI: jobs, hierarchy, breakpoints, brand

7. Sources and further reading

- Don Norman & Jakob Nielsen, "The Definition of User Experience (UX)" — Nielsen Norman Group (nngroup.com)
- Jakob Nielsen, "F-Shaped Pattern for Reading Web Content" — NN/g eyetracking research, 2006 and later replications
- Jakob Nielsen, "Jakob's Law of Internet User Experience," 2000
- W.E. Hick (1952) and R. Hyman (1953) — choice reaction time studies behind Hick's Law
- Paul Fitts (1954), "The information capacity of the human motor system"
- Jon Yablonski, Laws of UX (lawsofux.com) — plain-language summaries of all the laws above
- WCAG 2.2, Success Criterion 2.5.8 "Target Size (Minimum)"
- Your session pack: the Fig & Filter commission letter, the build handbook (with the specification and checklist as its appendices), and the content/ folder